Powers & Roots

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Powers & roots

What are powers (indices)?

- **Powers (or indices)** are the small 'floating' values that are used when a number is multiplied by itself repeatedly
  - 6^1 means 6
  - 6^2 means 6×6
  - 6^3 means $6 \times 6 \times 6$
- The big number at the **bottom** is called the **base**
- The small number that is **raised** is called the **index, power,** or **exponent**
- Any non-zero number to the **power of 0** is **equal to 1**
  - $3^0 = 1$
- Any number to the **power of 1** is **equal to itself**
  - $3^1=3$

What are square roots?

- **Roots** are the **reverse of powers**
- A **square root** of 25 is a number that when squared equals 25
  - The **two square roots** of 25 are 5 and -5
    - $5^2 = 25$ and $(-5)^2 = 25$
- Every **positive number** has **two square roots**
  - One is **positive** and one is **negative**
  - Negative numbers do not have a real square root
- The notation $\sqrt{}$ refers to the **positive square root** of a number
  - $\sqrt{25} = 5$
  - You can show both roots at once using the **plus or minus symbol** ±
  - Square roots of 25 are $\pm\sqrt{25} = \pm 5$

What are cube roots?

- A **cube root** of 125 is a number that when **cubed** equals 125
  - The cube root of 125 is 5
    - $5^3 = 125$
  - Unlike square roots, each number only has **one cube root**
  - Every **positive** and **negative number** has **one real cube root**

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

What are n^{th} roots?

- An ***n*th root** of a number is a value that when raised to the power *n* equals the original number
  - 3^5=243 therefore 3 is a 5th root of 243
- If *n* is **even**, there will be a **positive and negative** *n*th root
  - The 6th roots of 64 are 2 and -2
    - $2^6 = 64$ and $(-2)^6 = 64$
  - The notation $\sqrt[n]{}$ refers to the **positive *n*th root** of a number
    - $\sqrt[6]{64} = 2$
  - Negative numbers do not have a real *n*th root if *n* is even
- If *n* is **odd** then there will only be **one** *n*th root
  - The 5th root of -32 is -2
    - $(-2)^5 = -32$
  - Every **positive** and **negative** number will one real an *n*th root

How do I estimate a root?

- You can **estimate roots** by finding the **closest integer** roots
  - To estimate $\sqrt{20}$
    - We know that $\sqrt{16} = 4$ and $\sqrt{25} = 5$
    - So $\sqrt{20}$ must be between 4 and 5

What are reciprocals?

- The **reciprocal** of a number is **1 divided** by the number
  - The reciprocal of 2 is $\frac{1}{2}$
- A number and its reciprocal **multiply** together to give **1**
- You **switch** the numerator and denominator of a fraction to find its reciprocal
  - The reciprocal of $\frac{1}{4}$ is 4
  - The reciprocal of $\frac{3}{2}$ is $\frac{2}{3}$
- The **reciprocal** of a number can be written as an **index of -1**
  - 5^{-1} is the reciprocal of 5, so $\frac{1}{5}$
- This can be extended to other **negative indices**
  - 5^{-2} means the reciprocal of 5^2, so $\frac{1}{5^2}$ or $\frac{1}{25}$

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